Akashic Civics: Open-Source Tools for Criminal Justice Disparity Analysis & Civic Democracy

Akashic Civics Project Team

1 The Civic Problem

Local criminal justice decisions shape lives – but are rarely measured:

- Arrests, charges, convictions, and sentences happen at the local level
- District attorneys, sheriffs, and judges wield enormous discretion
- Disparity by race, income, and neighborhood is well-documented nationally – but local data is fragmented
- Without measurable baselines, communities cannot advocate for change or evaluate reform

2 What Akashic Civics Does

- Ingest local case data from any jurisdiction
- Run statistical tests (Z-tests, regression, causal inference)
- Track disparities over time with streaming analysis
- Export reports for advocacy, journalism, and policy
- Available in TypeScript, Python, and R

3 The Enhancement Roadmap (E16–E25)

E16 Data integration (FBI UCR, BJS) – TypeScript

E17 Advanced analytics (doubly-robust, Bayesian) – TypeScript

E18 Interactive CLI analysis wizard – TypeScript

E19 Jupyter Notebook interface – Python

E20 Publishing/release pipeline – JavaScript

E21 Extension system (plugins) – TypeScript

E22 R package (regression, causal, comparison) – R

E23 Streaming analysis with disparity alerts – TypeScript

E24 Web dashboard UI + Express API – TypeScript/JS

E25 Python statistical depth (NumPy/SciPy) – Python

4 Data Integration (E16)

- Pluggable DataSource interface
- FBIUCRSource and BJSSource implementations
- DataPipeline normalizes heterogeneous inputs into a common schema
- 9 passing tests

Implementation choice: We normalized upstream rather than asking users to clean data.

5 Advanced Analytics (E17)

- DoublyRobustEstimator – combines propensity scores with outcome regression
- BayesianHierarchicalModel – partial pooling across jurisdictions
- Effect sizes, confidence intervals, uncertainty quantification

Implementation choice: Doubly-robust methods protect against model misspecification.

6 Interactive CLI Wizard (E18)

- 5-step guided analysis for non-programmers
- Select data source, choose jurisdiction, select analysis type, configure parameters, execute and export

Implementation choice: A wizard rather than raw commands.

7 Jupyter + Python (E19 & E25)

- IPython magics and NotebookGenerator with comparison templates
- stats.py: descriptive stats, Z-tests, OLS, logistic regression, doubly-robust, disconfirmation, jurisdiction comparison

Implementation choice: Python wrappers try in-process NumPy/SciPy first, then fall back to CLI.

8 R Package (E22)

- CRAN-style package with descriptive stats, Z-tests, OLS/logistic regression
- Doubly-robust causal estimation, disconfirmation battery, multi-jurisdiction comparison
- R CMD check compliant, testthat tests, vignette

Implementation choice: R is the lingua franca of public policy and criminology.

9 Streaming Analysis (E23)

- StreamProcessor reads JSONL in chunks, never loads full dataset (1M+ cases)
- Incremental group statistics + disparity alerts when ratios cross thresholds
- Alert callback for external dashboards/alerts

Implementation choice: Bounded chunks + persistent manifest for resume and scalability.

10 Dashboard + API (E24)

- Express.js server with /api/health, /api/viz-specs, /api/analyze endpoints
- Static SPA serving disparity charts and KPI cards
- Disparity ratio, conviction rates by group, sentence length visualization

Implementation choice: Thin API layer over TypeScript statistical core.

11 Extension System + Publishing (E21 & E20)

- ExtensionRegistry discovers DataSource, Analysis, Export, Transformation plugins
- publish_pipeline.js: 11-step release cycle from lint to npm/PyPI/R
- 19 extension tests; dry-run mode for safety

12 Design Principles

1. Multi-language by design – Meet users where they are (TS, Python, R)
2. Statistical rigor first – No headline numbers without CI and effect sizes
3. Local data, local control – Data never leaves user's machine unless they choose
4. Composable, not monolithic – Use dashboard, CLI, or just the R package

13 Why Doubly-Robust? Why Bayesian?

- Simple rate comparisons → confounded by case mix
- Doubly-robust → consistent if either propensity or outcome model is correct
- Bayesian hierarchical → small jurisdictions get stable estimates through partial pooling

14 Why Stream, Why Alert?

- Batch analysis answers "what happened?" – Streaming answers "is it happening right now?"
- Alert thresholds create early warning for advocates and oversight bodies

15 Civic Impact – Decision Points to Local Leaders

Decision Point	**Who Decides**	**What Akashic Measures**
Charging	District Attorney	Conviction rate by demographic, controlling for severity
Bail / Pretrial	Judge	Pretrial detention disparity
Sentencing	Judge	Sentence length disparity, controlling for offense
Diversion	Prosecutor / Court	Disposition flow (convicted vs. dismissed vs. diverted)

16 A Concrete Scenario

Mid-sized county, 12,000 cases/year:

- Community loads 3 years of case data into Akashic Civics

- Doubly-robust: Black defendants convicted at 1.4× rate of White defendants ($p < 0.01$)
- Streaming alert: disparity ratio rose from 1.2× to 1.6× after new DA
- Community presents at hearing -¿ DA adopts new charging guidelines
- Next quarter: disparity drops to 1.1×

17 Promoting Democracy

- Transparency: Disparity metrics no longer locked in prosecutors' spreadsheets
- Participation: Community groups analyze the same data as officials
- Accountability: Elected leaders evaluated on measurable outcomes
- Reform evaluation: Did the policy actually reduce disparity? We can tell.

18 Get Involved

- Install: npm install akashic-civics or pip install akashic-civics-py or R CMD INSTALL
- Run: akashic wizard, akashic dashboard
- Contribute: extensions, data sources, visualizations, documentation
- GitHub: github.com/elijah/akashic-civics